INCH-POUND

MIL-M-27274E(USAF)
27 AUG 1993
MIL-M-27274D(USAF)
20 JUNE 1974

MILITARY SPECIFICATION

MASK, OXYGEN MBU-5/P

This specification is approved for use by the Departments of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of pressure breathing oxygen mask designated MBU-5/P.

1.2 Size. The oxygen masks shall be of the following sizes (see 6.5):

M27274-M10 - Long narrow	M27274-M30 - Regular wide
M27274-M20 - Regular narrow	M27274-M40 - Short narrow

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping Fiberboard
ZZ-R-765	Rubber, Silicone
BB-A-1034	Compressed Air, Breathing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Technology & Industrial Services Division, SA-ALC/TILDD, Kelly AFB, TX 78241-5609 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1660

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MILITARY

MIL-C-5040	Cord, Fibrous, Nylon
MIL-T-8363	Tape and Webbing, Textile, Woven, Nylon
MIL-O-27210	Oxygen, Aviator's Breathing, Liquid and Gas
MIL-V-27296	Valve, Oxygen Mask, Combination Inhalation and Exhalation

STANDARDS

FEDERAL

FED-STD-595	Colors
FED-STD-601	Rubber, Sampling and Testing

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-970	Standards and Specifications, Order of Precedence for Selection of
MIL-STD-147	Palletized Unit Loads
MIL-STD-831	Test Reports, Preparation of
MIL-STD-2073	DOD Material Procedures for Development and Application of Packaging Requirements
MS22064	Clamp, Oxygen Hose
MS27796	Connector-Bayonet, Three Pin, Oxygen Mask
MS27980	Fastener, Snap, Style 2 (Regular Wire Spring)
MS27983	Fastener, Snap, Style 4

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Air Force

60D3516	Face Form-Oxygen Mask, Size Short Narrow
60D3519	Face Form-Oxygen Mask, Size Regular Narrow
60D3522	Face Form-Oxygen Mask, Size Long Narrow
60D3525	Face Form-Oxygen Mask, Size Regular Wide
64D1676	Cover-Dust, MBU-5/P Oxygen Mask
7425180	Oxygen Mask-Type MBU-5/P - (A1)

(Application for copies should be addressed to SA-ALC/TILDE, Kelly AFB, TX 78241-5000.)

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2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents cited in the solicitation not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-624	Tear Resistance of Vulcanized Rubber
ASTM-D-1788	Standard Specification for Rigid Acrylonitrile-Butadiene-Styrene (ABS) Plastics
ASTM-D-3951	Packing, Commercial

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Materials. Materials shall conform to the applicable specifications and shall be as specified herein. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended. Materials that will come in contact with oxygen shall be entirely free from oil, grease, or any other combustible materials.

3.3.1 Silicone material. Silicone rubber shall be used in the facepiece, the delivery tube, and the microphone cable guides of the oxygen mask and shall conform to Class 3a, Grade 50 of ZZ-R-765, except that the minimum brittle point shall be -56°C and the Shore A hardness shall be 45 ±5. It shall be smooth and free from flash, foreign materials, or imperfections. The silicone rubber shall be post-cured in a well ventilated forced air oven. The date (by year and quarter) the silicone rubber was cured shall be molded on the facepiece and the delivery tube of the mask.

3.3.2 Plastic materials. The hardshell shall be fabricated by injection molding or forming from acrylonitrile-butadiene-styrene molding plastic conforming to ASTM-D-1788. The color of the hardshell shall be the same as the color of the mask facepiece. The thickness of the hardshell shall be 3/64 ±1/64.

3.3.3 Webbing. The hardshell webbing shall conform to Type I of MIL-T-8363. The color of the webbing shall be substantially the same as the color of the hardshell.

3.3.4 Nylon cord. The nylon cord shall conform to Type IA of MIL-C-5040.

3.3.5 Protective treatment. When materials that are subject to corrosion in oxygen, salt air, or any other atmospheric conditions likely to occur during service usage are used in the construction of the mask, they shall be protected against corrosion in the manner that in no way prevents compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

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3.4 Design and construction. The design and construction of the mask shall be in accordance with Air Force drawings 60D3516, 60D3519, 60D3522, 60D3525 and 7425180 (see figures 1-9 inclusive). Air Force drawings 60D3516, 60D3519, 60D3522, and 60D3525 should be used as general guidelines in the design of tooling.

3.4.1 Plug. Each mask shall have a suitable leak tight plug for the microphone flange fabricated from metal, plastic, or hard rubber.

3.4.2 Antifreeze chamber. The antifreeze chamber and the body of the mask shall be of one-piece construction as shown (see figures 4 and 5). A 1/16 inch hole located as shown in figures 4 and 5 shall vent the antifreeze chamber.

3.4.3 Combination valve. A combination inhalation-exhalation valve conforming to MIL-V-27296, Type I, shall be inserted in the mask as shown (see figure 5).

3.4.4 Delivery tube. The delivery tube for the mask shall be constructed of corrugated silicone as shown (figure 6). The delivery tube shall have one upper cable guide (see figure 7) and located as shown (see figure 1).

3.4.5 Clamp. The delivery tube shall be attached to the combination valve by means of a hose clamp conforming to Scott Aviation Division of Figgie, Intl. Inc., Sierra Madre, part number 450-134A or approved equal. Clamp assembly pliers conforming to Scott Aviation, part number 450-801 may be used to attach the clamp.

3.4.6 Connector. A connector conforming to MS27796 shall be attached to the delivery tube by means of a hose clamp conforming to MS22064-5 or other approved hose clamps. A lower cable guide (see figure 7) shall cover this hose clamp and shall be located as shown (see figure 1).

3.4. 7 Limited stretch cord. A nylon cord and snap assembly as shown (see figure 8) shall be installed inside the delivery tube by attaching a nylon loop to the combination valve retention pin and a metal snap to the connector retention pin. The snap shall be Pacific Marine Supply Company, Seattle, Washington, size No. 2/0 (1 3/16 inches in length, 110-pound test), or equal, and shall be attached to one end of the nylon cord as shown (see figure 8).

3.4.8 Hardshell assembly. The hardshell assembly shall consist of the hardshell including the installed hardshell webbing, harness fasteners with screws, and suspension harness restrained. The hardshell assembly shall be as shown (see figures 1 and figure 9).

3.4.8.1 Hardshell. The hardshell for the mask shall closely follow the contour of the outside of the facepiece and shall be designed to assure that all of the performance requirements specified herein are met. The edges of the hardshell nearest the user's face shall be flared, and all sharp edges shall be rounded. Dimensions given (see figures 2 and 3) may be used as general guidelines in the design of the hardshell.

3.4.8.2 Hardshell webbing. The webbing shall be riveted to the hardshell as shown (see figure 1) by 1/8 inch tubular rivets having a minimum head diameter of 9/32 inch.

3.4.8.3 Harness fasteners. Fasteners conforming to MS27980 and MS27983 shall be located in the hardshell as shown (see figure 9). Two machine screws (3-56NF-3A by 0.375 inch long) shall secure the harness assembly to the hardshell. These machine screws shall be provided as part of the hardshell assembly.

3.4.8.4 Suspension harness restrainer. A suitable device as shown (see figure 1) shall be provided for restricting the lateral movement of the harness assembly when the mask is used with a continuous single strap suspension harness assembly. The device shall be provided as a component of the hardshell assembly and shall be approved by the procuring activity.

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3.4.9 Dust cover. The mask shall be equipped with a dust cover to prevent dust, dirt, and foreign objects from entering the mask cavity when the mask is not in use. The cover shall be made of durable, nonbrittle, lightweight plastic material contoured to rest around the face edge of the hardshell and to be retained on the hardshell with an elastic webbing around the front edge of the hardshell at the microphone contour. The configuration of the cover shall conform to 64D1676. The cover shall not rest on the face piece contour between the seal area and the body of the facepiece.

3.5 Performance.

3.5.1 Mask leakage. The inboard leakage shall not exceed 100 milliliters per minute when tested (see 4.6.2).

3.5.2 Combination valve installation, inlet performance (low suction). The total leakage through the combination valve inlet portion shall be 15 milliliter per minute or less when tested (see 4.6.3).

3.5.3 Combination valve installation, inlet performance (high suction). The total leakage through the combination valve inlet portion shall be 150 milliliters per minute or less when tested (see 4.6.4).

3.5.4 Combination valve installation, exhalation performance. A pressure of 20 millimeters of mercury in the facepiece and a pressure between 15 and 19.9 millimeters of mercury in the inlet tubing shall cause the exhalation portion of the valve to open when tested (see 4.6.5).

3.5.5 Combination valve installation, exhalation resistance. The exhalation resistance shall not be greater than 4.67 millimeters of mercury with a flow of 100 liters per minute when tested (see 4.6.6).

3.5.6 Delivery tube leakage. The delivery tube shall not leak when subjected to an internal air pressure of five pounds per square (see 4.6.7).

3.5. 7 Delivery tube strength. The delivery tube shall not tear or separate when tested (see 4.6.8).

3.5.8 Tear resistance. The silicone rubber shall have a minimum tear resistance of 150 pounds per inch when tested (see 4.6.9).

3.5.9 Hardness. The Shore-A-Durometer hardness of the silicone rubber shall be 45 ±5 when tested (see 4.6.10).

3.5.10 Toxicity. The silicone rubber shall not contain any chemicals or materials that have toxic properties (see 4.6.11).

3 .5.11 Subjective use. The silicone rubber shall not produce any objectionable odors, shall not be tacky, and shall not have any other properties that cause discomfort or affect wear ability. The mask shall have no detrimental or objectionable performance characteristics when evaluated (see 4.6.12).

3.5.12 Hardshell flexibility. The mask hardshell shall withstand a decrease in width (as measured by dimension B of figure 2) of 1 1/2 inches without visible cracking or fracture when subjected to compressive loading (see 4.6.13).

3.6 Color. The color of the hardshell, webbing, and all silicone rubber used in the oxygen mask assembly shall be green, approximately matching color number 34079 conforming to FED-STD-595.

3. 7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

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3.8 Identification of product. The mask assembly shall be marked with the following specific information conforming to MIL-STD-130. The Federal Stock Number shall be omitted from the nameplate.

Mask, Oxygen: MBU-5/P
Mask Size: M27274-*
Manufacturer: *
Date of Manufacture: (Quarter and Year)*

3.8.1 Tubing. The mask end of the tubing shall be identified by molding "Mask End" into the rubber. The date of manufacture (quarter and year) shall be molded on either end of the tubing.

3.9 Workmanship. The mask shall be constructed in accordance with good commercial practice, and shall be free from any defects which might affect appearance or function. The mask shall be clean and dry, free from oil, grease, and other foreign matter. All metal and plastic parts shall be smooth and free of sharp edges. All excess flash shall be properly trimmed. The facepiece and its related parts shall be smooth and free from blisters, porous areas, cracks, abrasions, bubbles, and foreign material.

3.10 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions.

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77° ±18°F, and at a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from these values, proper allowance shall be made for the change instrument reading.

4.3.2 Gas. The gas used in testing shall be oxygen conforming to type I of MIL-O-27210. The gas used in all other testing shall be oxygen conforming to type I of MIL-O-27210, water pumped nitrogen conforming to type I, class 1, grade B of BB-N-411, or compressed air conforming to grade A or C of BB-A-1034. If nitrogen or air is used, appropriate density correction factors shall be applied to the flow meter used.

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4.4 First article inspection. First article inspection shall be preformed on all performance requirements IAW paragraph 3.5. This inspection shall include the examinations and tests as described herein (see 4.6) before regular production is started.

4.4.1 First article. When specified in the contract, three complete first articles shall be submitted to the procuring activity at the time the first article test report is submitted for approval. The Government reserves the right to perform any or all tests, inspections and examinations specified herein.

4.4.1.1 Inspection facility. The first article shall be inspected in a laboratory acceptable to the Government or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity. The required number of first articles shall be stated in the contract. The inspections shall be as described (see 4.6).

4.4.2 Test report. In the event that the first article testing is performed at the approved contractor's laboratory, contractor shall, after completion of the first article tests, prepare three copies in accordance with MIL-STD-831 of the first article test report and silicone rubber report containing the following information (see 6.3):

a. Compound number and name of manufacturer.

b. Composition of compound, including fillers and catalysts, giving parts by weight of each constituent.

c. Mixing and curing instructions.

d. Any additional information the manufacturer desires to furnish for better identification of the material or to show that the material has desirable characteristics for oxygen mask usage.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual tests.

b. Sampling plans and tests.

4.5.1 Individual tests. Each oxygen mask shall be subjected to the following tests as described (see 4.6).

a. Examination of product.

b. Mask leakage.

c. Combination valve installation, inlet performance (low suction).

d. Combination valve installation, exhalation performance.

e. Delivery tube leakage.

4.5.2 Sampling plans and tests.

4.5.2.1 Sampling plan A. Two masks of each size specified in the contract or order shall be selected at random from each lot of each size of 1,000 masks (or fraction thereof) and shall be subjected to the following tests as described (see 4.6):

a. Combination valve installation, inlet performance (high suction).

b. Combination valve installation, exhalation resistance.

c. Delivery tube strength.

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4.5.2.2 Sampling plan B. One mask facepiece of each size specified in the contract or order shall be selected at random from each batch of silicone rubber compound received from the manufacturer. Test sheets fabricated (see 4.4.1) shall also be made from each batch. The test sample shall be subjected to the following tests as described (see 4.6):

a. Tear resistance.

b. Hardness.

4.5.2.3 Sampling plan C. If hardshells are molded in an essentially continuous operation, three hardshells shall be selected at random from each day's production of each mold cavity and subjected to the test described (see 4.6.13). If hardshells are not molded in an essentially continuous operation, three mask hardshells of each size of mask specified in the contract or order (fabricated from each batch of material received from the manufacturer and molded under the same conditions and at substantially the same time) shall be selected at random and subjected to the test described (see 4.6.13).

4.5.3 Reject criteria. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined and corrected. Individual tests may be continued pending the investigation of a sampling test failure. However, final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of this specification. The investigation of a test failure could indicate that defects may exist in items already accepted; if so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Inspection methods.

4.6.1 Examination of product. Each oxygen mask shall be carefully examined to determine conformance to the requirements of this specification with respect to materials, construction, workmanship, dimensions, and marking.

4.6.2 Mask leakage. Each mask (with combination valve installed correctly and the microphone plug in place) shall be placed on a suitable test stand and adjusted so there is no leakage around the face seal. The inboard leakage shall not exceed 100 milliliters per minute when a suction of 13 millimeters of mercury is applied within the facepiece of the mask with the free end of the mask tubing plugged.

4.6.3 Combination valve installation, inlet performance (low suction). The inlet valve portion shall be completely dry throughout this test. The inlet valve shall be inspected for imperfect seating of the flapper on the seat and leaks around the valve seat by applying and maintaining a suction of 0.19 millimeters of mercury at the free end of the delivery tube. The total leakage through the combination valve inlet portion shall not be more than 15 milliliters per minute. If necessary, a suitable means shall be used to prevent collapse of the hose.

4.6.4 Combination valve installation, inlet performance (high suction). This test is the same as 4.6.3 except that a suction of 22.4 millimeters of mercury shall be used. The leakage through the combination valve inlet portion shall not exceed 150 milliliters at this suction.

4.6.5 Combination valve installation, exhalation performance. Each mask shall be placed on a suitable test stand and all suction and pressure from the previous test exhausted. A pressure of 20 millimeters of mercury shall be applied at the mask tubing. The pressure at the mask tubing shall then be gradually reduced, causing the inlet valve to close and causing a pressure differential to be applied across the exhalation valve. The mask exhalation valve shall open with a pressure in the tube between 15 and 19.9 millimeters of mercury. If an extreme pressure is encountered, the inlet valve shall be rechecked for leakage. If the valve is properly seated and the mask still fails to function properly, the valve shall be replaced and the test repeated.

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4.6.6 Combination valve installation, exhalation resistance. The mask (with the combination valve installed) shall be placed on a suitable test stand. At an exhalation flow of 100 liters per minute, the pressure inside the mask shall not exceed ambient pressure by more than 4.67 millimeters of mercury.

4.6. 7 Delivery tube leakage. One end of the delivery tube shall be plugged and an air pressure of 5 pounds per square inch shall be applied to the other end. The tube shall be submerged just under the surface of water. Bubbles rising from the hose after 10 to 15 seconds of submergence shall be cause for rejection.

4.6.8 Delivery tube strength. One end of the tubing shall be attached to the combination valve mounted on the mask body. The other end shall be connected to a fitting simulating a connector conforming to MS27796. Clamps for the connections shall be of the type specified herein. One of the end connections shall be attached to a support. A weight of 63 pounds ± 1 pound shall be attached to the other connection so that a straight pull is exerted on the tubing for a period of 3 seconds ± 2 seconds. The tubing shall not tear or separate from the end connections. The nylon anti stretch cord shall be attached during this test. Upon completion of this test the delivery tube shall be subjected to the test of (see 4.6.7).

4.6.9 Tear resistance. The tear resistance of the test sheets and sections of rubber cut from the sealing area of the facepiece shall be determined in accordance with ASTM-D-624-54, die B. Minimum tear resistance shall be 150 pounds per inch.

4.6.10 Hardness. The hardness of the test sheets and sections of rubber cut from the sealing of the facepiece in both the nose and check areas shall be determined in accordance with method 3021 of FED-STD-601, and shall have a shore A hardness of 45 ±5.

4.6.11 Toxicity. Toxicity shall be determined from the chemicals used in formation of the silicone rubber compound.

4.6.12 Subjective use. Objectionable odors, tackiness, and objectionable or detrimental performance characteristics shall be determined by subjective evaluation. The mask shall be used by human subjects at altitudes from ground level to 50,000 feet, at temperature from -40°F to + 160°F or any combination of such altitude and temperature for periods up to 2 hours. Subjective use tests are optional tests to be conducted at the discretion of and by the procuring activity when the mask appears to have performance characteristics or other properties which might be detrimental or objectionable for human use.

4.6.13 Hardshell flexibility. The hardshell shall be placed between the heads of a suitable compression testing machine with the heads contacting the sides of the hardshell at the point of maximum width. The hardshell shall be oriented so that dimension B (see figure 2) is parallel to the direction of movement of the heads. A suitable means of measuring the decrease in hardshell width under load shall be provided. The hardshell shall be deformed until failure or until the required decrease in width is attained, whichever occurs first. The rate of head travel shall be from 0.20 to 0.25 inch per minute. The hardshell shall withstand a decrease in width of at least 1 1/2 inches without visible cracking or breaking.

4.6.13.1 Special tool. A special compression tool for applying the load to the hardshell shall be used to minimize any effects caused by shifting of the test machine heads. This tool shall be constructed so that loading is truly parallel to dimension B (see figure 2) and is applied through tool surfaces that are flat, parallel, and perpendicular to dimension B (see figure 2).

4.7 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or commercial as specified (see 6.2).

5.1.1 Level A. Masks shall be preserved to the MIL-STD-2073-1A requirements for this level.

5.1.2 Level C. Masks shall be preserved to the MIL-STD-2073-1A requirements for this level.

5.1.3 Commercial. Masks shall be preserved in accordance with the applicable requirements of ASTM-D-3951.

5.2 Packing. Packing shall be level A, B, C or commercial as specified (see 6.2).

5.2.1 Level A. Masks that have been preserved and packaged as specified (see 5.1) shall be packed in exterior-type shipping containers that conform to PPP-B-601, overseas type. The closure of the shipping container shall be in accordance with the appendix of the applicable shipping container specification.

5.2.2 Level B. Masks that have been preserved and packaged as specified (see 5.1) shall be packed in exterior-type shipping containers that conform to PPP-B-636, class weather resistant. The closure of the shipping container shall be in accordance with the appendix of the applicable shipping container specification.

5.2.3 Level C. Masks that have been preserved and packaged as specified (see 5.1) shall be packed in accordance with the requirements of MIL-STD-2073-1A.

5.2.4 Commercial. Masks that have been preserved and packaged as specified (see 5.1) shall be packed in accordance with the requirements of ASTM-D-3951.

5.3 Marking.

5.3.1 Levels A, B, and C. In addition to any special or other identification markings required by the contract (see 6.2), each unit pack and exterior container shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Masks shall be marked in accordance with the applicable requirements of ASTM-D-3951.

5.4 Palletization. Unitized loads, commensurate with the level of packing specified in the contract or order shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern shall first be approved by the contracting officer.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The oxygen mask covered by this specification is for use in dispensing gaseous oxygen from pressure demand regulators to aircrew members engaged in high altitude flying.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

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c. Size or sizes required (see 1.2).

d. When first article is required (see 3.1, 4.4.1 and 6.4).

e. Levels of preservation, packaging, and packing required (see section 5).

f. When palletization is required (see 5.4).

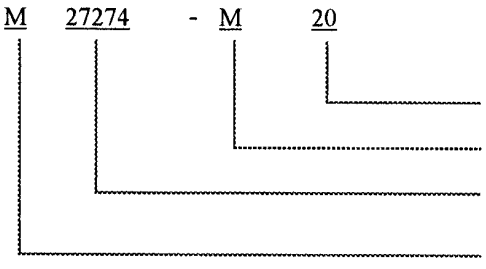
6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title
4.4.2	DI-T-2072	Inspection Test Reports

The above DIDs were those cleared as of the date of this inspection. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4.1. The contracting officer should also include specified instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Part or Identifying Number (PIN). The PIN system is as follows:



Example of PIN: M27274-M20

Size if applicable (see table I)

Item (see table I)

Specification number

An item defined by inch-pound units

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TABLE I. Items and size

Items	
F	Facepiece
H	Hardshell, Mask
M	Mask Assembly
01	Tube
03	Cord and snap
05	Cable guide upper
07	Cable guide lower
09	Clamp upper

Size	
10	Long narrow
20	Regular narrow
30	Regular Wide
40	Short narrow

6.6 Subject term (key word) listing.

Air Crew Survival
Long Narrow
Regular Narrow
Regular Wide
Reports, Test
Rubber Silicone
Short Narrow

6. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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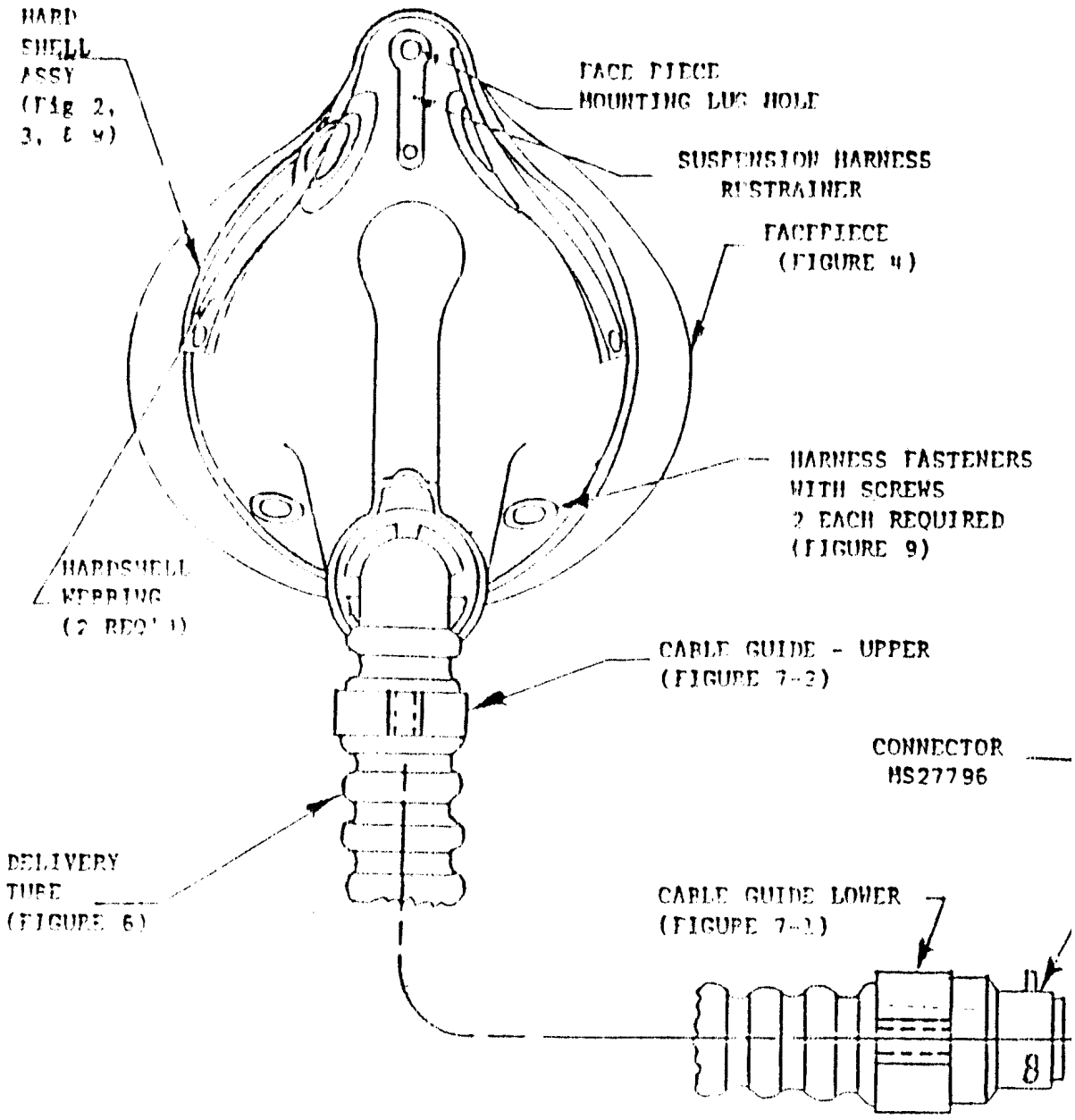
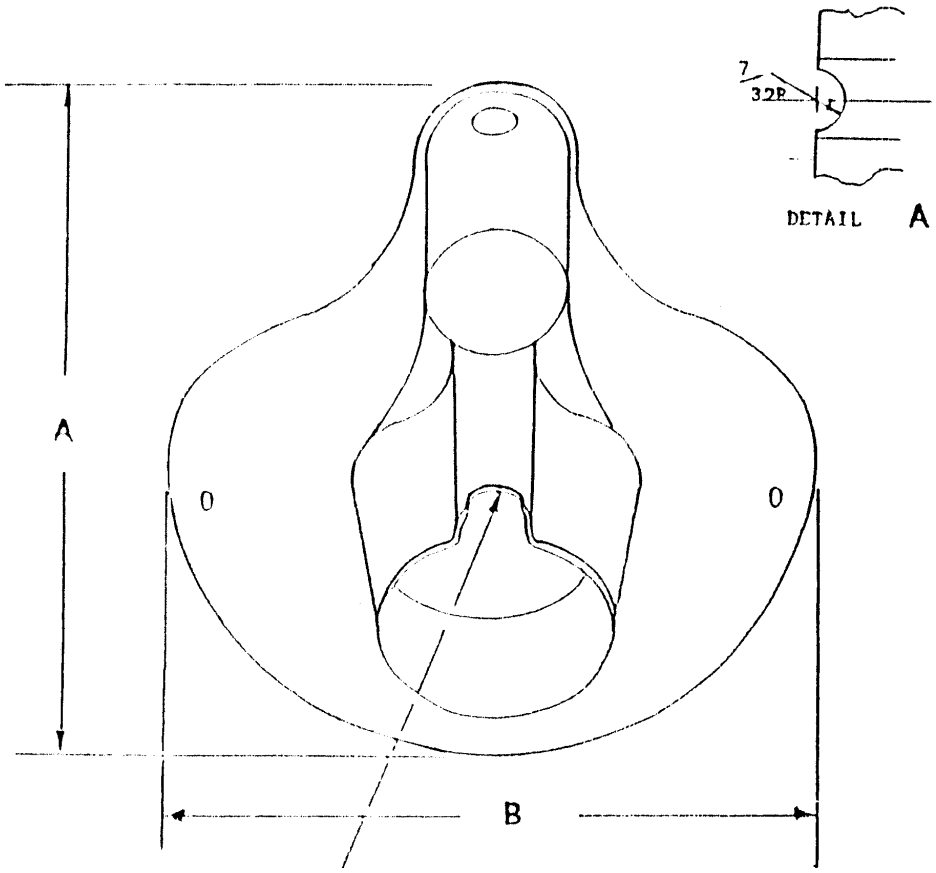


FIGURE 1. Mask assembly.

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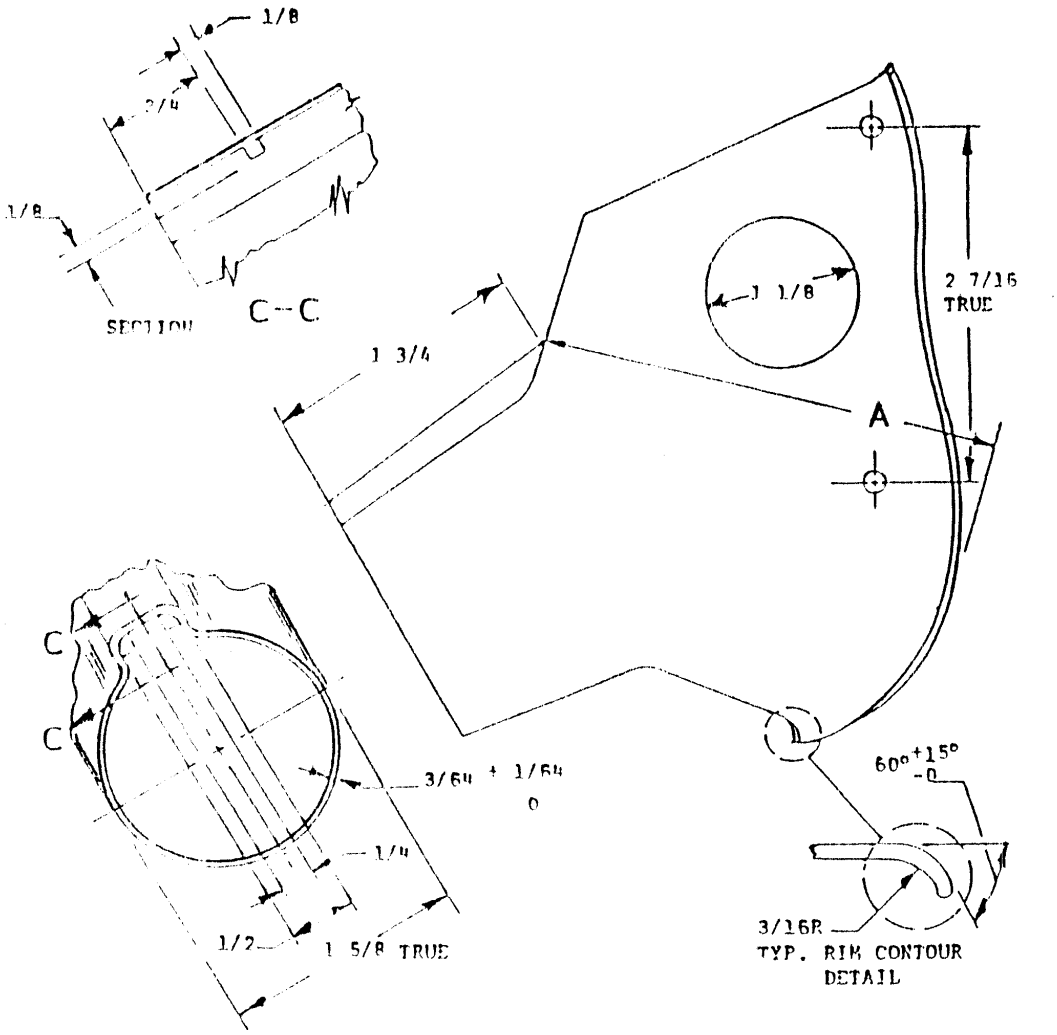
CUTOUT TO ALLOW
MICROPHONE WIRE TO BE
ATTACHED
(SEE DETAIL "A")

DIMENSIONS IN
INCHES

DIMENSIONS	SHORT NARROW	REGULAR NARROW	LONG NARROW	REGULAR WIDE
A	4 13/16	4 7/8	5 1/8	4 3/4
B	4 9/16	4 1/4	4 3/16	4 3/8

FIGURE 2. Mask hardshell.

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DIMENSION	SHORT NARROW	REGULAR NARROW	LONG NARROW	REGULAR WIDE
A	3 1/6	3 1/8	2 15/16	2 5/8

DIMENSIONS IN INCHES

FIGURE 3. Mask hardshell details.

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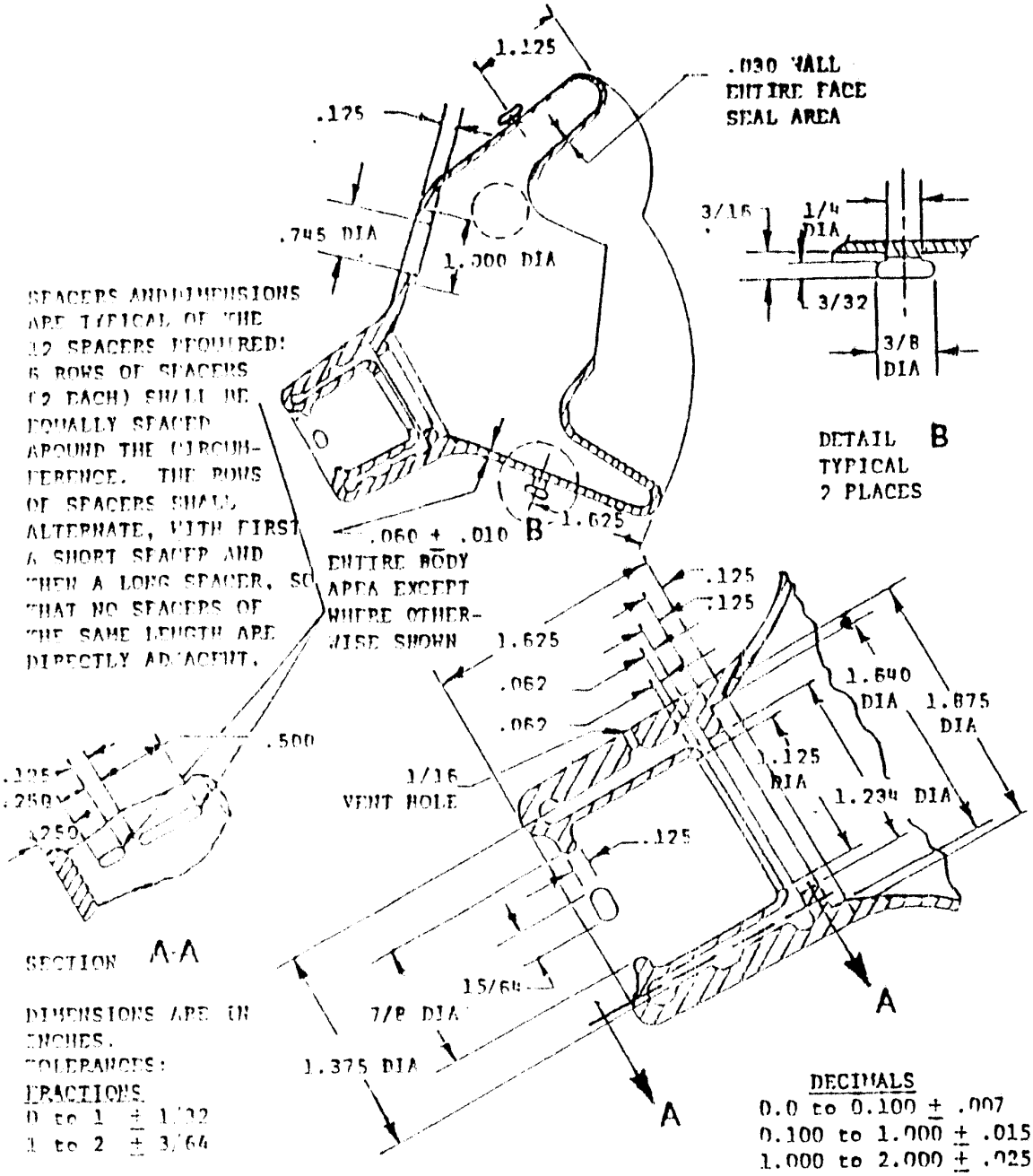


FIGURE 4. Fixed dimensions of the facepiece.

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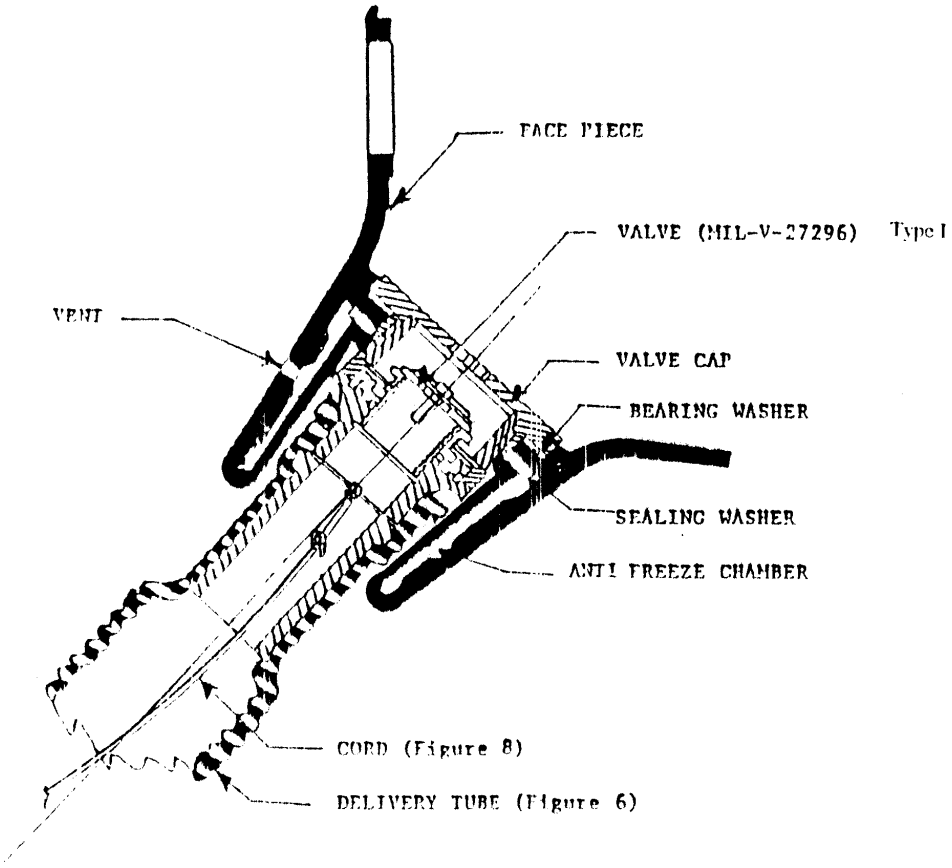
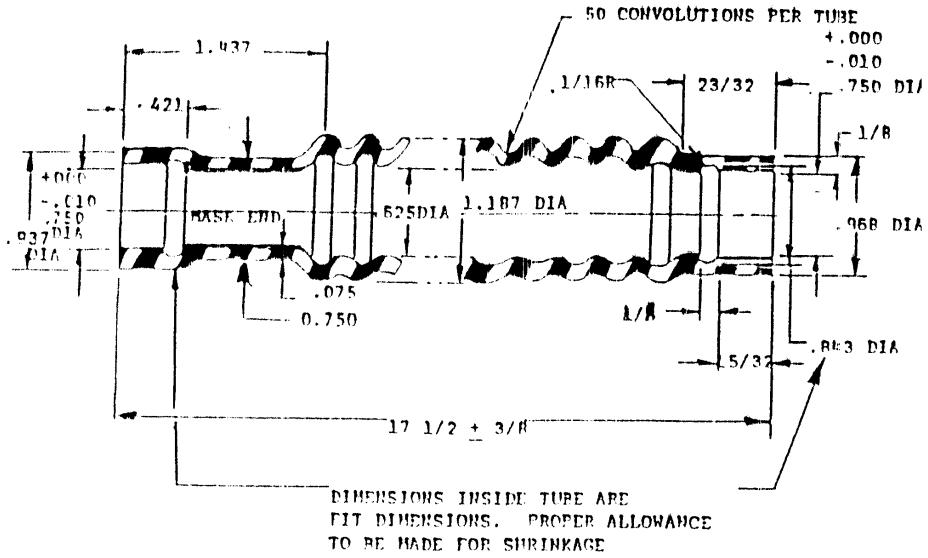


FIGURE 5. Valve installation.

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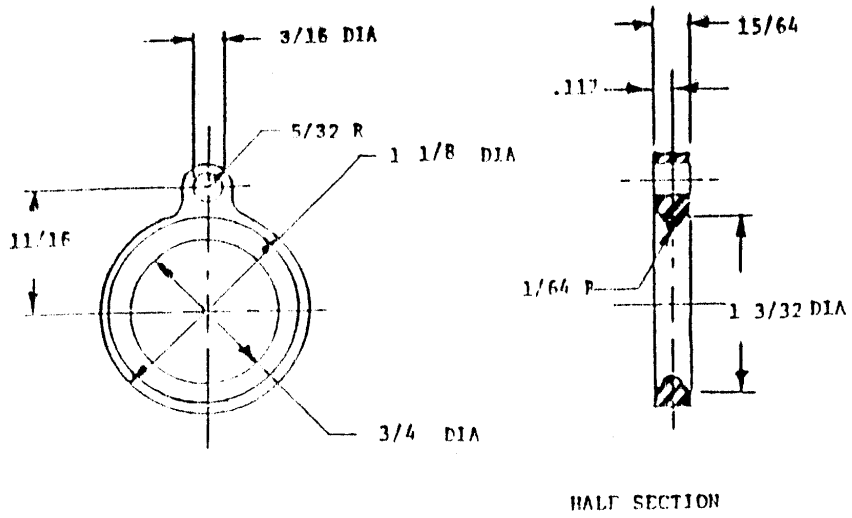


DIMENSIONS IN INCHES

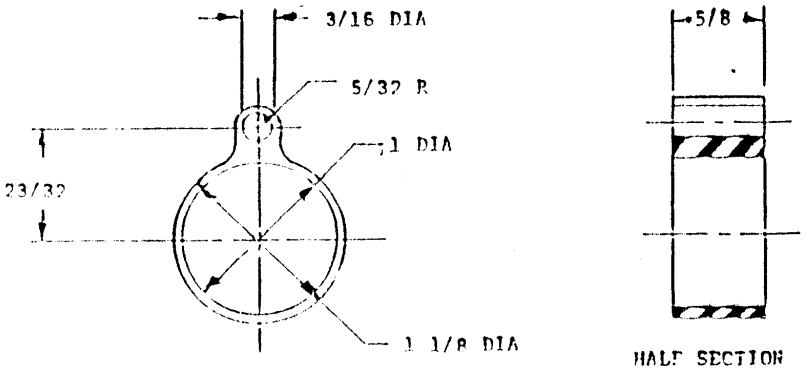
TOLERANCE ON FRACTIONS	DECIMALS
± 1/32	± .010

FIGURE 6. Delivery tube.

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Cable guide upper



DIMENSION IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCE FOR ALL DIMENSIONS
SHALL BE ± 1/64

Cable guide lower

FIGURE 7. Cable guides.

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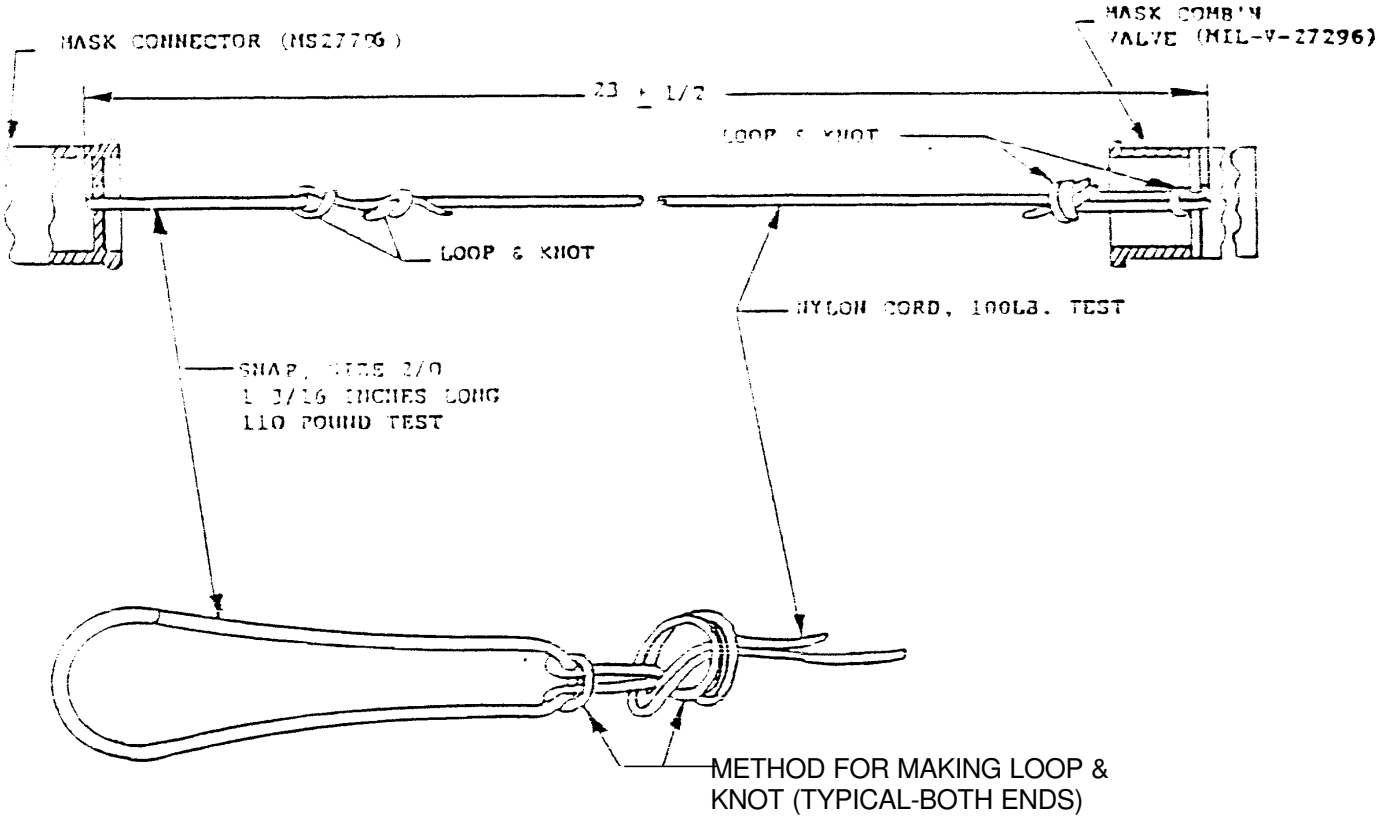
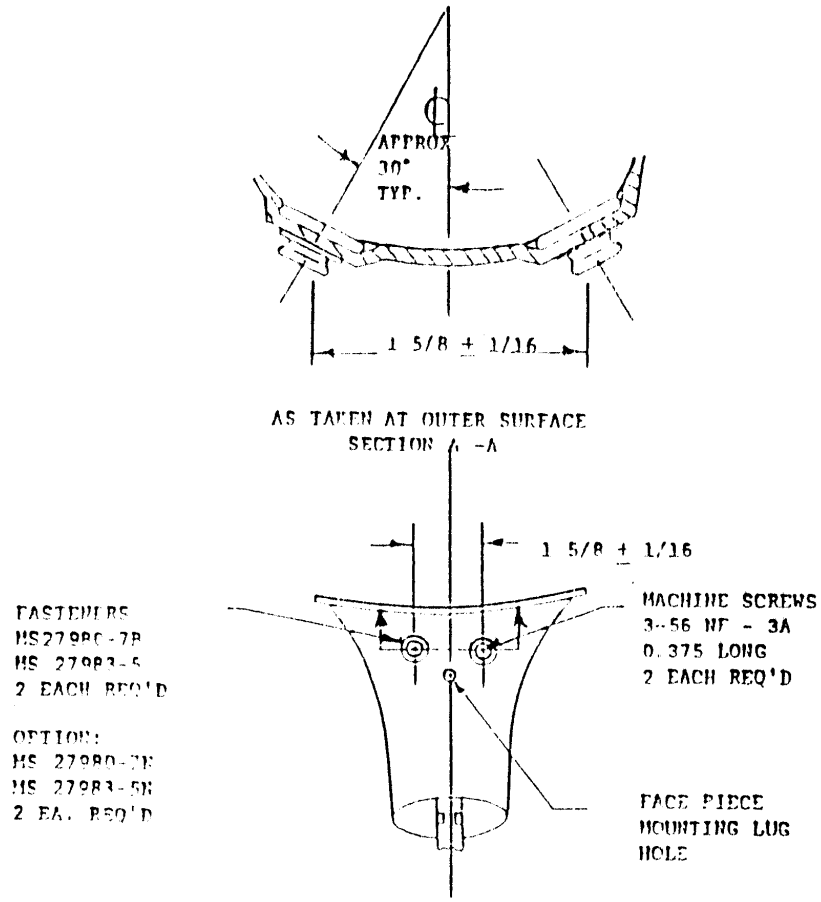


FIGURE 8. Nylon cord and snap assembly.

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DIMENSIONS IN INCHES

FIGURE 9. Harness fastner arrangement.

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Custodian
Air Force - 99

Preparing activity:
Air Force -82

(Project 1660-F541)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-27274E	27 August 1993

3. DOCUMENT TITLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
SA-ALC/TILDD	512-925-6277
c. ADDRESS (Include Zip Code) Kelly AFB, TX 78241-5609	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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